
XData /sql

User Guide & Handbook

Automated SQL Query Grading Platform

A complete, screenshot-based guide for
Administrators, Instructors and Students

First start in 30 seconds: run `docker compose up --build`, open <http://localhost>,
and sign in with the seeded administrator account `admin1 / admin1`.

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1 Introduction

XData-Web is a web platform for learning SQL and grading SQL queries *automatically*. Instead of comparing query *text*, it checks whether a student's query is *semantically equivalent* to the instructor's reference solution, and — even when it is not — awards *partial credit* based on how close the query structure is (tables, columns, conditions, joins, grouping, ...).

This handbook walks through every part of the application, organised by the three user roles:

- **Administrator** — manages users and courses for the whole system.
- **Instructor** — creates courses, schemas, assignments and questions, and monitors student progress.
- **Student** — solves SQL questions and receives instant feedback.

Two additional tools are available to everyone who is signed in: the **SQL Lab** (a grading simulator) and the **Dataset Playground** (a test-data generator).

2 Getting started

2.1 Starting the application

The only prerequisite is **Docker**. From the project root, run:

```
docker compose up --build
```

This starts the database, the backend and the frontend. On the first start the database schema and a default administrator are created automatically, so there is no manual database setup. When the logs settle, open:

- **The app:** <http://localhost>
- **API docs (Swagger):** <http://localhost:8080/swagger-ui.html>

2.2 First-time initialisation

On a fresh installation exactly **one** account exists — the administrator. **Initialise the system by signing in with** `admin1 / admin1`.

Recommended first steps after the initial login:

1. Change the administrator password (key icon, top right).
2. Create a course (Admin Center → *Courses*).
3. Create at least one instructor and assign them to the course.
4. Let the instructor add a schema, an assignment and questions.
5. Create students and assign them to the course.

2.3 Signing in

Open the app and enter your user ID and password. New users receive their credentials from an administrator; the administrator starts with `admin1/admin1` (see §2.2).

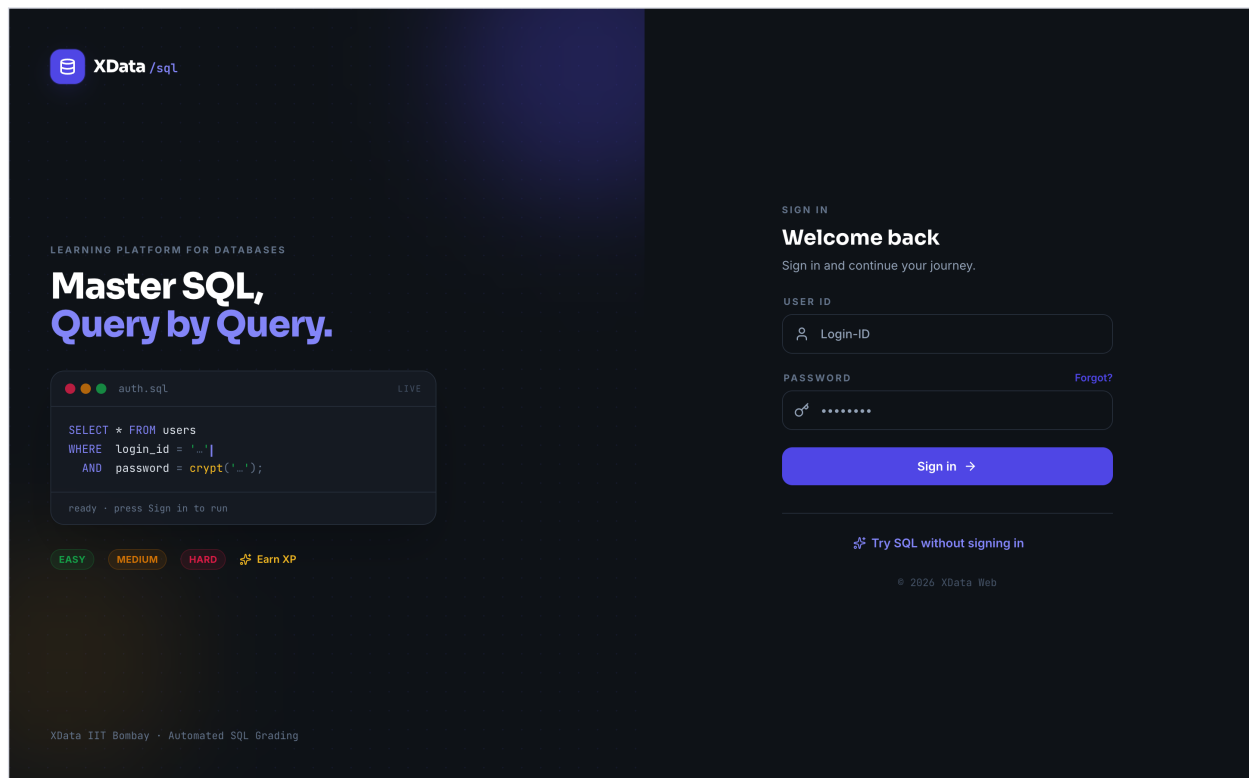


Figure 1: The login screen. Sign in with your user ID and password.

The link “*Try SQL without signing in*” opens a public SQL sandbox — handy for trying the editor without an account.

3 Administrator guide

After signing in as an administrator you land in the **Admin Center**, which has two tabs: *Users* and *Courses*.

3.1 Creating users

The *Users* tab contains the **Create new user** form on the left and the list of existing users on the right.

The screenshot displays the XData Admin Center interface. At the top, there's a navigation bar with 'XData /sql', 'Dashboard', 'SQL Lab', and 'Dataset Playground'. The main header shows 'Admin Center' with the subtitle 'System-wide management & monitoring.' and tabs for 'Users' and 'Courses'. Below this, the 'User management' section includes buttons for 'Template' and 'Import CSV'. The 'Create new user' form is prominent, featuring fields for 'FULL NAME' (e.g. John Doe), 'LOGIN ID' (e.g. jdoe), 'EMAIL ADDRESS' (john@example.com), 'PASSWORD' (masked), and 'ROLE' (STUDENT). A 'COURSE ASSIGNMENT' section shows 'Datenbanken I' selected. A 'Create user' button is at the bottom right of the form. Below the form is a table titled 'Assigned users' with 5 total users. The table has columns for STATUS, NAME, LOGIN ID, COURSE, ROLE, CREATED, and ACTIONS. The users listed are: schülerrr (STUDENT, 6/21/2026), Erika Lehrer (INSTRUCTOR, 6/20/2026), Max Schüler (STUDENT, 6/20/2026), Daniel Instructor (INSTRUCTOR, 6/17/2026), and Admin (ADMIN, 6/17/2026). Each row has a set of action icons (copy, delete, edit, share, trash).

ADMINISTRATION
Admin Center
System-wide management & monitoring.

USERS
User management

Create new user

FULL NAME: e.g. John Doe
LOGIN ID: e.g. jdoe
EMAIL ADDRESS: john@example.com
PASSWORD: *****
ROLE: STUDENT
COURSE ASSIGNMENT: ☒ Datenbanken I

Create user

Assigned users 5 TOTAL

Q Name, login ID or email... Newest first

STATUS	NAME	LOGIN ID	COURSE	ROLE	CREATED	ACTIONS
ACTIVE	S schülerrr	schülerrr	DB1-2026	STUDENT	6/21/2026	[Icons]
ACTIVE	E Erika Lehrer	Lehrer1	DB1-2026	INSTRUCTOR	6/20/2026	[Icons]
ACTIVE	M Max Schüler	schueler1	DB1-2026	STUDENT	6/20/2026	[Icons]
ACTIVE	D Daniel Instructor	daniel	No course	INSTRUCTOR	6/17/2026	[Icons]
ACTIVE	A Admin	admin1	No course	ADMIN	6/17/2026	[Icons]

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Figure 2: Admin Center → Users: the user-creation form and the user list with per-row actions.

To create a single user:

1. Fill in **Full name**, **Login ID** (used to sign in) and **Email**.
2. Set an initial **Password**.
3. Choose a **Role**: STUDENT, INSTRUCTOR or ADMIN.
4. Tick one or more courses under **Course assignment**.
5. Click **Create user**. The new account appears in the list immediately.

The screenshot shows the 'Admin Center' interface for user management. The 'Create new user' form is the central focus, with fields for Full Name, Login ID, Email Address, Password, Role, and Course Assignment. The 'Assigned users' table below shows a list of users with columns for Status, Name, Login ID, Course, Role, Created, and Actions.

STATUS	NAME	LOGIN ID	COURSE	ROLE	CREATED	ACTIONS
ACTIVE	Anna Beispiel	anna1	Datenbanken I	INSTRUCTOR	2024-10-26	[Icons for actions]

Figure 3: The create-user form filled in for a new instructor, with a course ticked.

Roles in short: STUDENTS solve questions; INSTRUCTORS manage courses, schemas, assignments and grading; ADMINS manage all users and courses.

3.2 Importing many users at once (CSV)

For whole classes, use **Import CSV** (top right of the Users tab). Click **Template** first to download a correctly-formatted CSV, fill in one row per user, then upload it with **Import CSV**. All users are created in one step.

3.3 Managing existing users

Each row in the user list offers the following actions:

- **Disable / Enable** — block or restore sign-in without deleting the account.
- **Sign in as this user** — *impersonate* the user to see exactly what they see (great for support and for verifying an instructor's setup). A confirmation dialog appears first.
- **Edit** — change name, email, role or course assignment.
- **Reset password** — set a new password for the user.
- **Delete** — permanently remove the account.

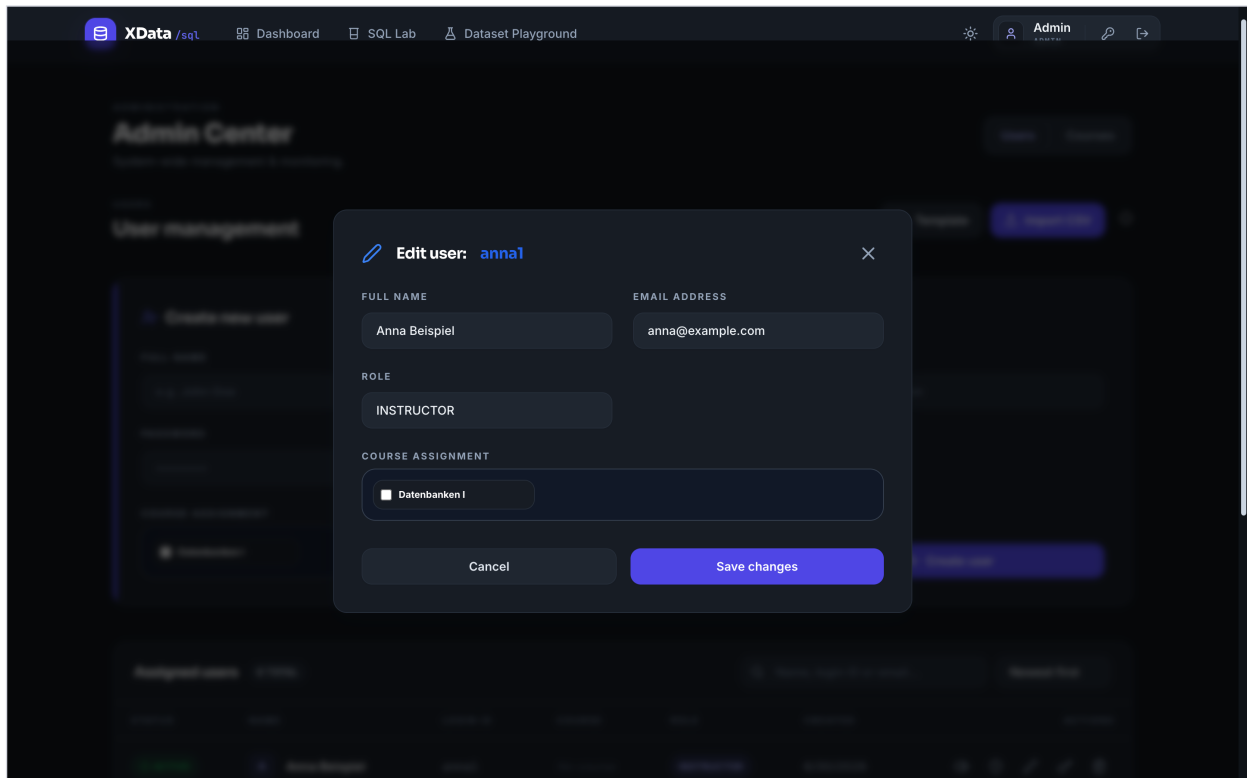


Figure 4: The Edit-user dialog: change name, email, role and course assignment.

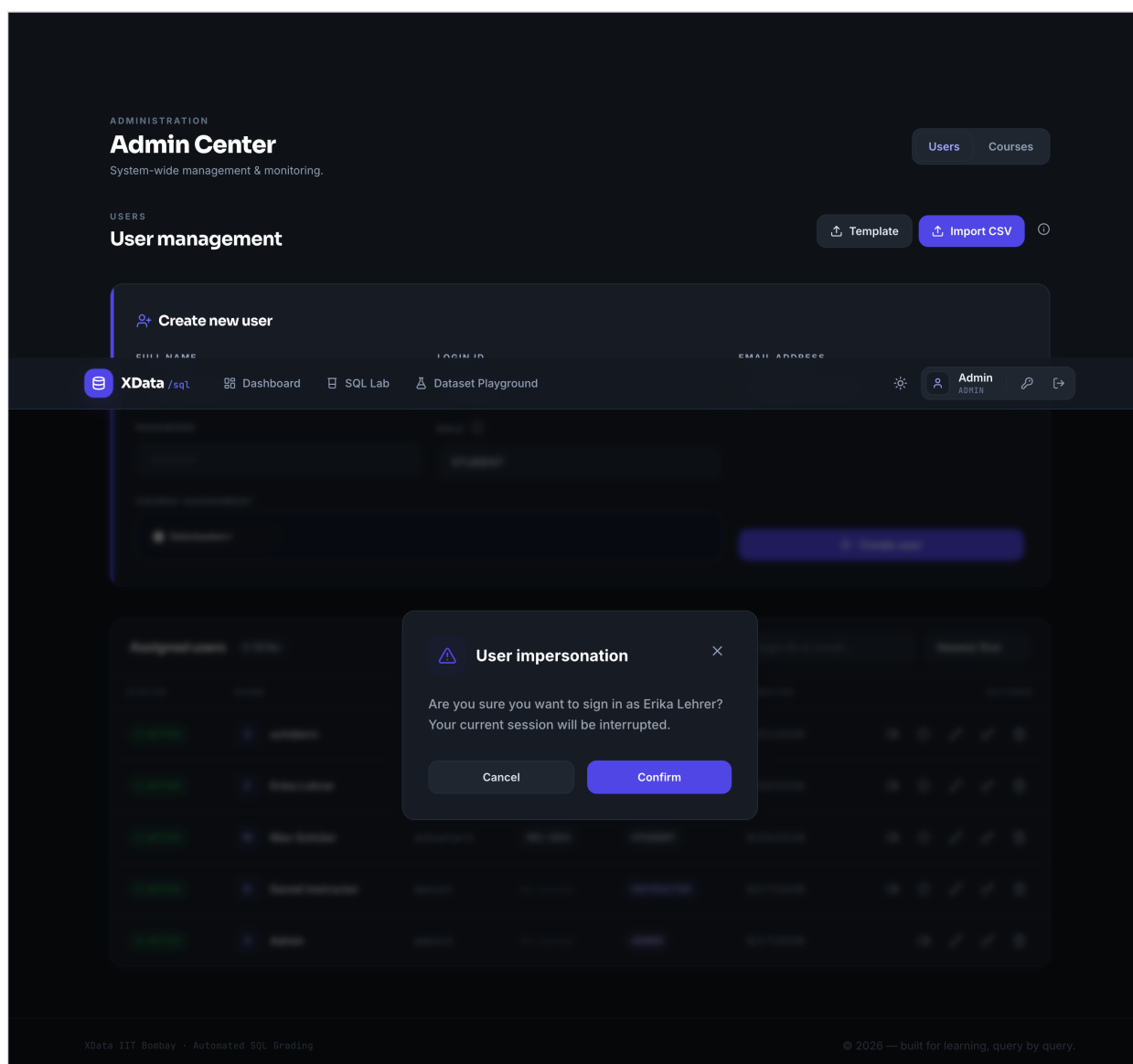


Figure 5: “Sign in as this user” asks for confirmation before impersonating — your admin session is replaced by the target user’s view.

3.4 Managing courses

The *Courses* tab lists all courses and lets you create new ones. A course bundles its instructors, students, schemas and assignments.

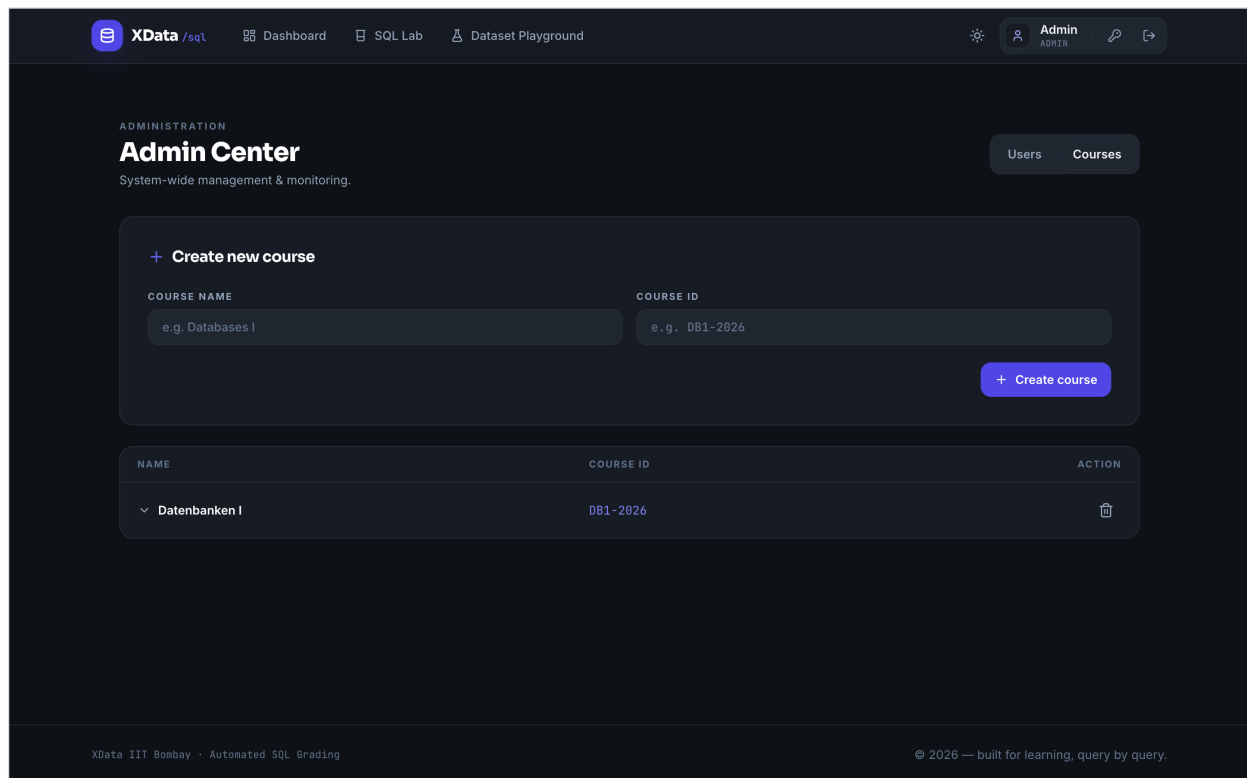


Figure 6: Admin Center → Courses: create and manage courses.

4 Instructor guide

Instructors work in the **Instructor Panel**. The left sidebar groups every tool; the *Advanced* section holds the setup tools (course materials, SQL schemas, databases, courses).

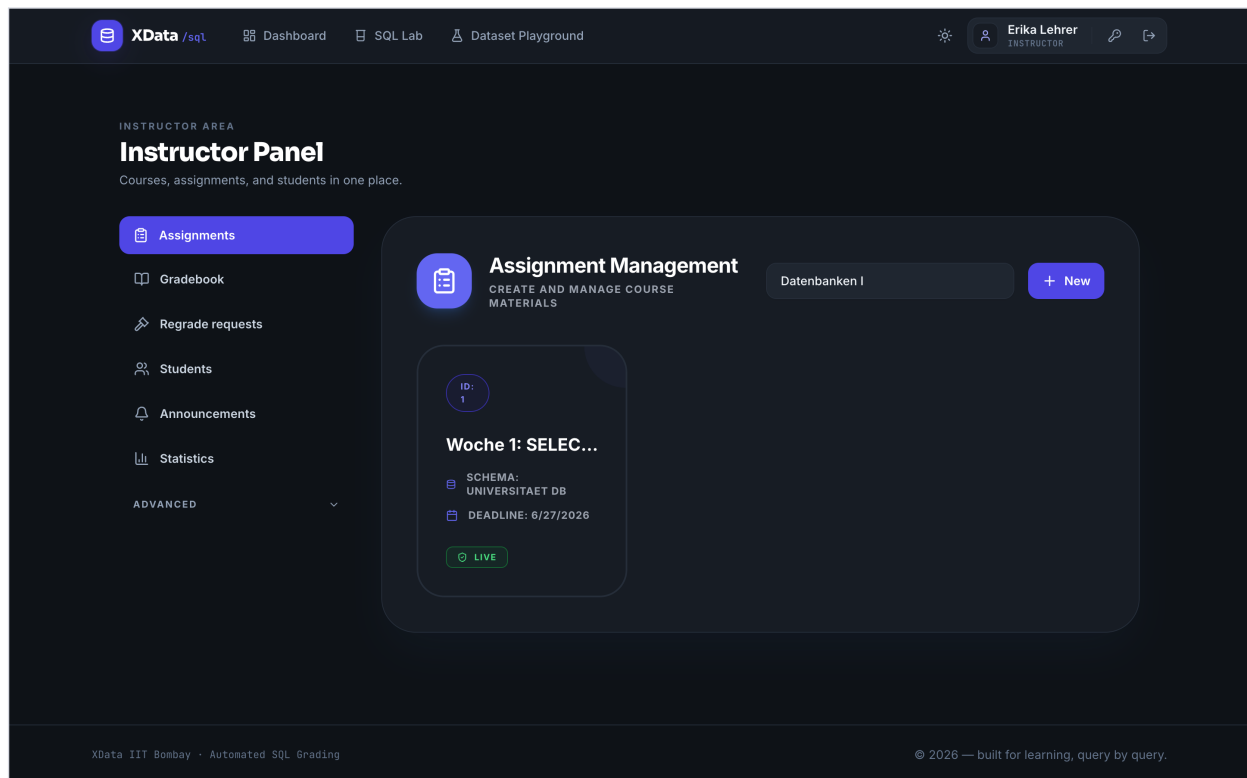


Figure 7: The Instructor Panel. The sidebar gives access to assignments, gradebook, students, statistics and the advanced setup tools.

4.1 Defining schemas

Before creating questions, define the database schema the questions run against. Open *Advanced* → *SQL Schemas*, select the course, and add the schema (table definitions). Schemas are reused by assignments, the SQL Lab and the Dataset Playground.

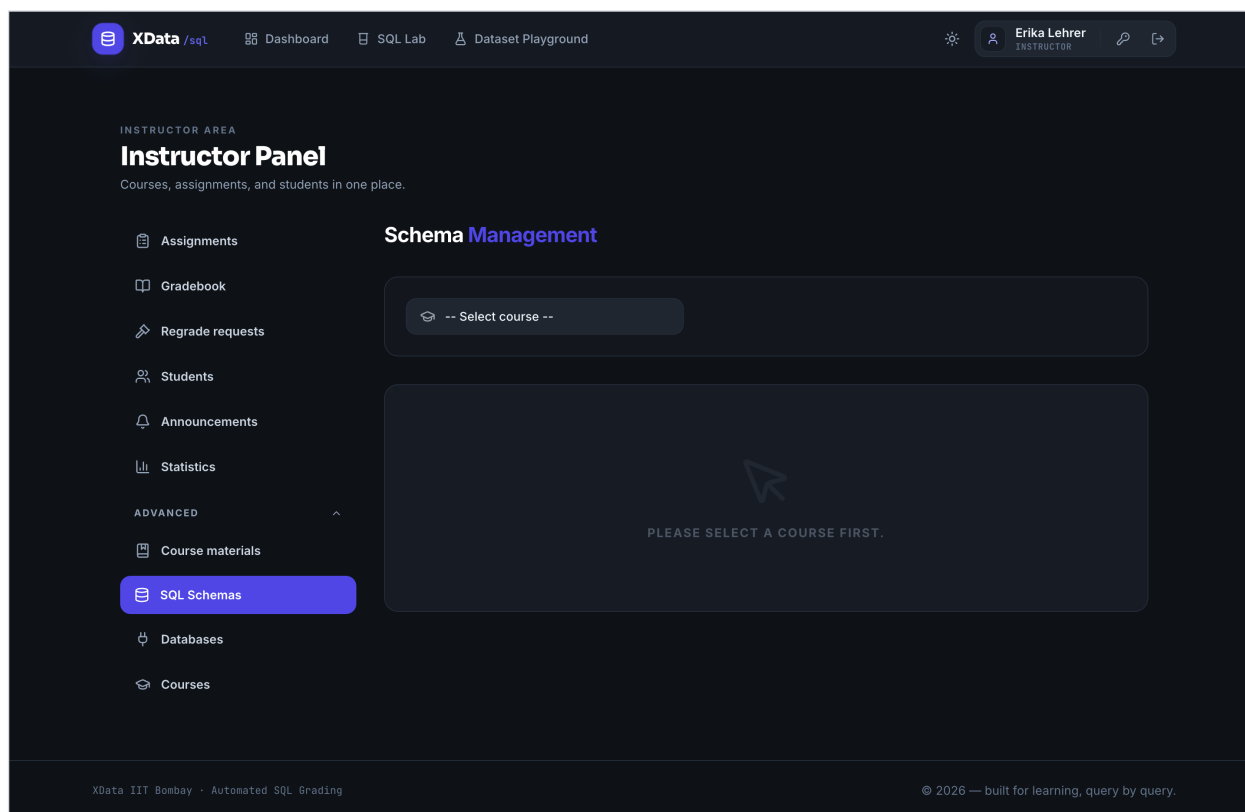


Figure 8: Schema management. Select a course, then create or edit the SQL schema used by its questions.

4.2 Assignments and questions

Open *Assignments*, pick the course, and click **New** to create an assignment. For each assignment you set a **schema** and a **deadline**, and add **questions**. Every question has a problem statement, a reference (model) solution, a difficulty, point value and optional hints. From an assignment you can also **Edit**, **Duplicate**, view **Statistics**, grant **Deadline extensions**, or **Delete** it. Set an assignment to *LIVE* to make it visible to students.

4.3 Monitoring progress — the Gradebook

The *Gradebook* shows every student's score per assignment, plus a total, and can be exported as **CSV**.

INSTRUCTOR AREA

Instructor Panel

Courses, assignments, and students in one place.

Assignments

Gradebook

Regrade requests

Students

Announcements

Statistics

ADVANCED

Course materials

SQL Schemas

Databases

Courses

GRADING

Gradebook

Datenbanken I (DB1-2026) [CSV](#)

STUDENT	WOCHE 1: SELECT BASICS	TOTAL
Max Schüler	10/10	100%
schülerrrr	0/10	0%

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Figure 9: The Gradebook: per-student, per-assignment scores with a CSV export.

4.4 Other instructor tools

- **Regrade requests** — review and resolve student requests to re-check a submission.
- **Students** — see the roster of the selected course.
- **Announcements** — post messages to a course.
- **Statistics** — per-assignment analytics (open them via the chart icon on an assignment).
- **Databases** — register external database connections used for grading.
- **Course materials** — attach links and resources to a course.

5 Student guide

5.1 The learning dashboard

After signing in, a student sees their personal dashboard: level and XP progress, a daily streak, a **leaderboard**, earned **badges**, and the list of **assignments**.

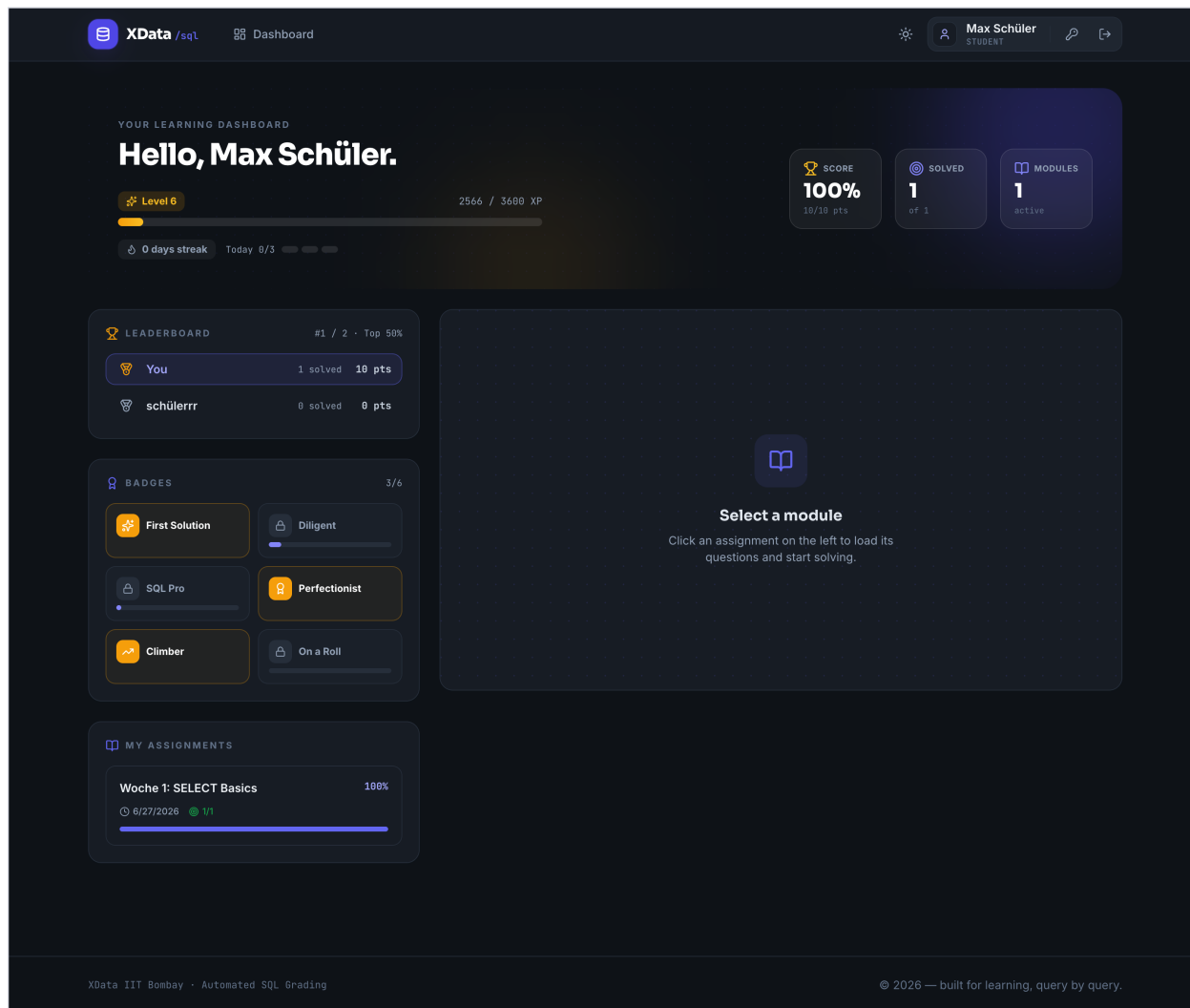


Figure 10: The student dashboard: XP and level, leaderboard, badges and assigned modules.

5.2 Opening an assignment

Click an assignment under *My assignments* to load its questions. Each question shows its difficulty, tags and point value; solved questions are marked with a green check.

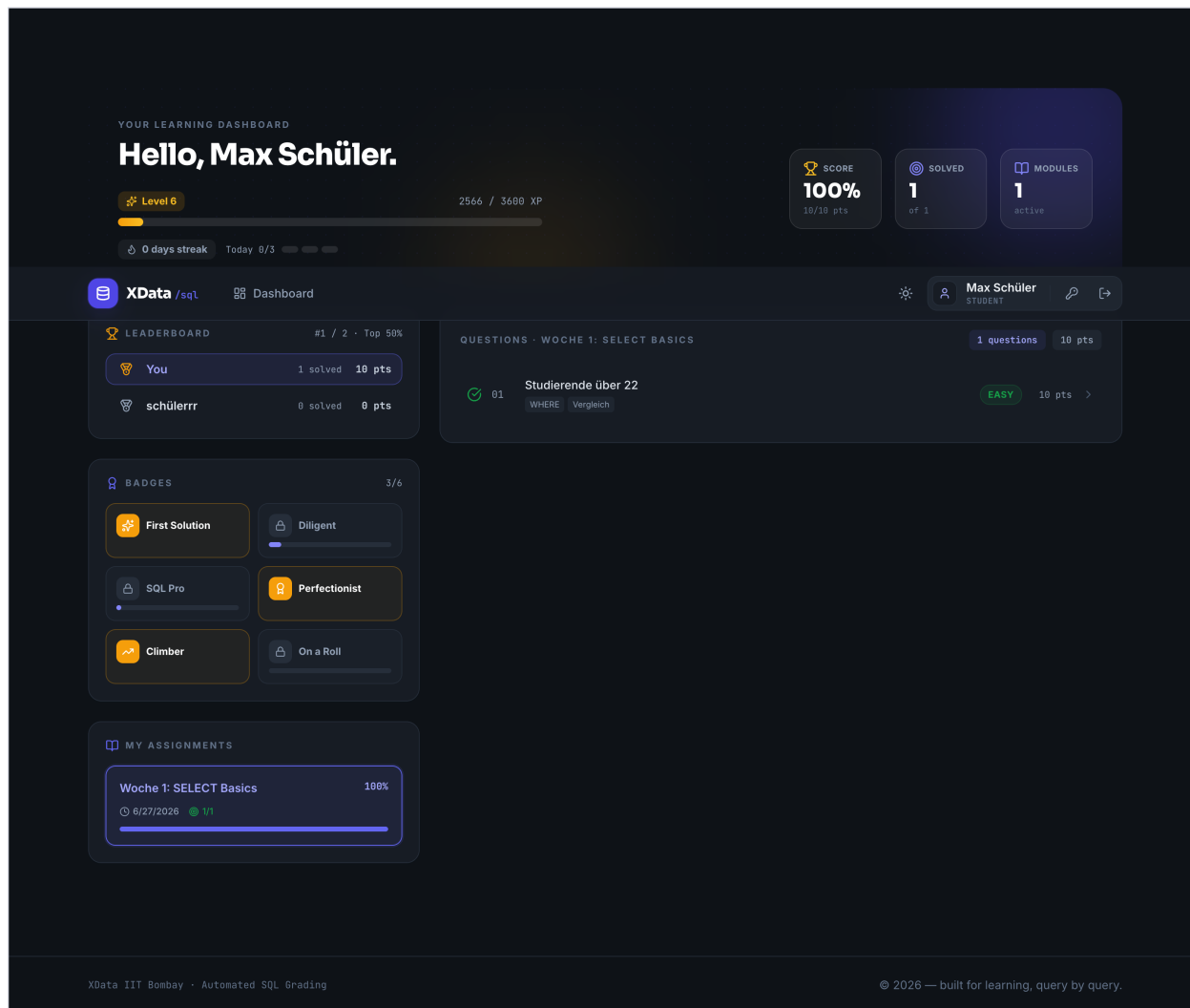


Figure 11: Selecting an assignment loads its question list on the right.

5.3 Solving a question

Open a question to reach the solving view. It contains:

- The **problem statement**, plus **Schema**, **Cheat sheet** and **Sample data** helpers.
- Step-by-step **hints** (optional — revealing them may cost points).
- The **SQL editor** with syntax highlighting and auto-completion.
- **Run** (try the query) and **Submit answer & check** (grade it).

YOUR LEARNING DASHBOARD

Hello, Max Schüler.

Level 6 2544 / 3400 XP

0 days streak Today 9/3

SCORE 100% **SOLVED 1** **MODULES 1**

LEADERBOARD #1 / 2 - Top 50%

You 1 solved 10 pts

schülerrrr 0 solved 0 pts

BADGES 3/6

First Solution Diligent

SQL Pro Perfectionist

Climber On a Roll

MY ASSIGNMENTS

Woche 1: SELECT Basics 100%

8/27/2028 9/1

QUESTIONS - WOCHE 1: SELECT BASICS 1 questions 10 pts

Studierende über 22 EASY 10 pts

WHERE Vergleich

PROBLEM STATEMENT

Gib die Namen aller Studierenden aus, die älter als 22 Jahre sind.

HINTS Show hint (0/2)

Stuck? Reveal hints step by step.

1

Run Submit answer & check

LAST RESULT 100%

OVERALL SCORE 3.0 / 3.0 100%

COMPONENT	YOUR SOLUTION	MODEL SOLUTION
Tables (FROM)	✓ STUDENTS (+1.0)	STUDENTS
Columns (SELECT)	✓ name (+1.0)	name
Conditions (WHERE)	✓ age > 22 (+1.0)	age > 22
Joins (JOIN)	-	-
Grouping (GROUP BY)	-	-
Sorting (ORDER BY)	-	-
Filter (HAVING)	-	-
Duplicates (DISTINCT)	✓ No (-0.0)	NO

Compare expected output Request regrade

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Figure 12: The question view: problem statement, hints, SQL editor, and the structural result with per-component feedback.

5.4 Understanding the result

After submitting, the **Last result** panel shows an overall score and a **component-by-component** comparison of your query against the model solution — tables, columns, conditions, joins, grouping, sorting, filters and duplicates. Each matching part earns partial credit, so a near-miss still scores. You can **Compare expected output** or, if you believe the grade is wrong, **Request a regrade** (the instructor reviews it under *Regrade requests*).

6 SQL Lab (Diagnostic Lab)

The **SQL Lab** is a grading *simulator*. Pick a schema, enter a model solution and a candidate query, optionally adjust the grading weights, and run a structural evaluation — the same engine that grades student submissions. It is ideal for instructors testing a question or for anyone exploring how the partial marking works.

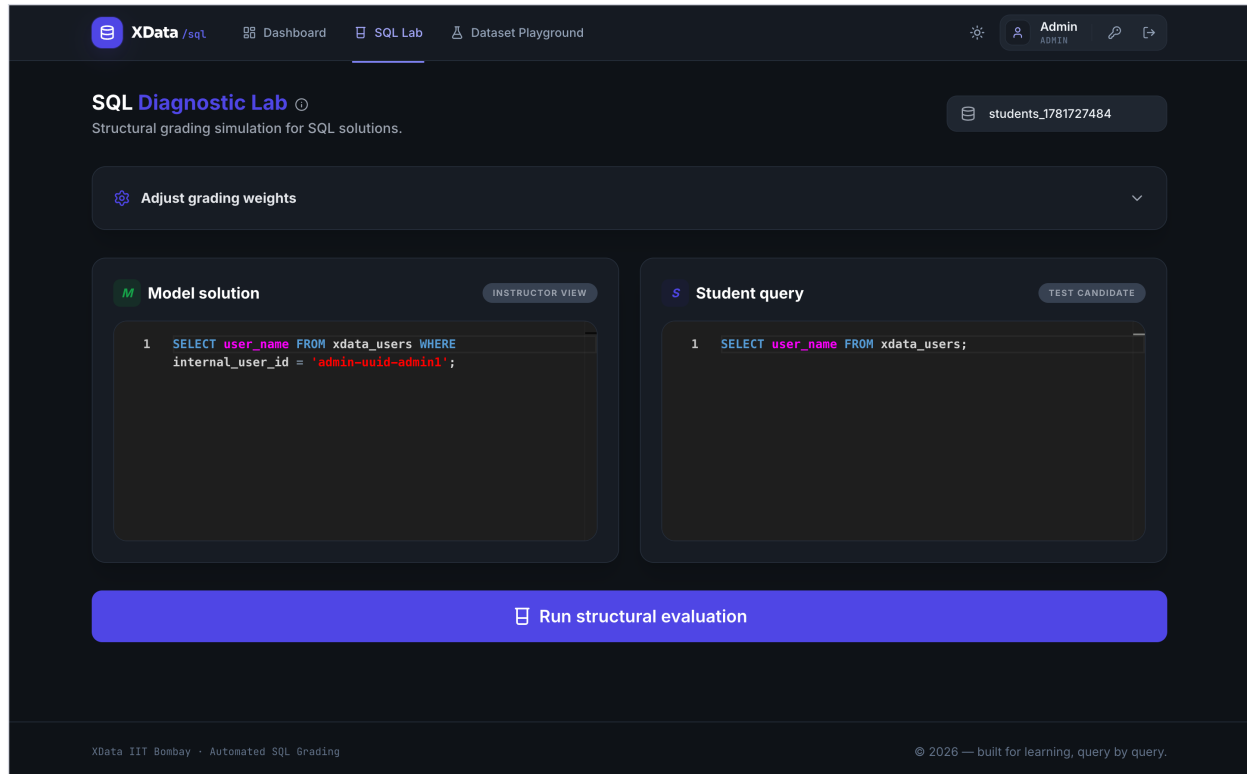


Figure 13: SQL Lab: a model solution (left) and a student/candidate query (right).

The screenshot displays the SQL Diagnostic Lab interface. At the top, there's a navigation bar with 'XData /sql', 'Dashboard', 'SQL Lab', and 'Dataset Playground'. The user is logged in as 'Admin'. The main heading is 'SQL Diagnostic Lab' with a subtitle 'Structural grading simulation for SQL solutions.' and a student ID 'students_1781727484'. Below this is a section to 'Adjust grading weights'. The interface is split into two panels: 'Model solution' (Instructor View) and 'Student query' (Test Candidate). The 'Model solution' shows a query: `1 SELECT user_name FROM xdata_users WHERE internal_user_id = 'admin-uuid-admin1';`. The 'Student query' shows a query: `1 SELECT user_name FROM xdata_users;`. A large blue button labeled 'Run structural evaluation' is positioned below the queries. The 'STRUCTURAL ANALYSIS & FEEDBACK' section shows an 'OVERALL SCORE' of 2.0 / 3.0 (67%). Below this is a table comparing the 'MAIN QUERY' components between 'YOUR SOLUTION' and 'MODEL SOLUTION'.

COMPONENT	YOUR SOLUTION	MODEL SOLUTION
Tables (FROM)	✓ XDATA_USERS (+1.0)	XDATA_USERS
Columns (SELECT)	✓ user_name (+1.0)	user_name
Conditions (WHERE)	-	internal_user_id = 'admin-uuid-admin1'
Joins (JOIN)	-	-
Grouping (GROUP BY)	-	-
Sorting (ORDER BY)	-	-
Filter (HAVING)	-	-
Duplicates (DISTINCT)	✓ No +0.0	NO

At the bottom, there's a footer with 'XData IIT Bombay · Automated SQL Grading' and '© 2026 — built for learning, query by query.'

Figure 14: SQL Lab result: an overall score plus a per-component structural breakdown.

7 Dataset Playground

The **Dataset Playground** generates targeted test datasets that expose errors (“mutants”) in a SQL query, using the SMT solver (Z3). Select a schema, paste a reference query, choose which **mutation types** to target, and click **Generate dataset**. The generated INSERT statements appear on the right and can be copied or downloaded as .sql.

Figure 15: Dataset Playground: choose a schema and mutation types, then generate a discriminating dataset.

Important — qualify your columns. The reference query must use *table-qualified* column names, e.g. `SELECT employees.name FROM employees WHERE employees.salary > 50000` (not just `name/salary`). This is a requirement of the underlying data-generation engine.

A successful run produces a dataset that includes *boundary* values — exactly the rows needed to tell a correct query apart from a subtly wrong one. For `employees.salary > 50000`, the engine generates rows with `salary = 50001` (passes), `50000` (the boundary that kills a `>=` mutant) and a non-matching value:

The screenshot displays the XData Dataset Playground interface. At the top, there's a navigation bar with links to XData /sql, Dashboard, SQL Lab, and Dataset Playground. The main header shows the 'Dataset Playground' title and a 'Load schema (.sql)' button. Below this, the interface is divided into two main sections.

Left Panel:

- SELECT DATABASE SCHEMA:** A dropdown menu showing 'Company DB'.
- REFERENCE QUERY (SQL):** A text area containing the query: `1 SELECT employees.name FROM employees WHERE employees.salary >`.
- MUTATION TYPES:** A section with checkboxes for various mutation types:
 - ☒ Selection
 - ☒ Joins (Equi)
 - ☐ Joins (Non-Equi)
 - ☒ Aggregation
 - ☐ Distinct
 - ☐ Group By
 - ☐ Having
- Generate dataset:** A blue button at the bottom right of the left panel.

Right Panel:

- GENERATED DATASET (SQL INSERTS):** A text area showing the generated SQL inserts for the query. The inserts are:


```
insert into DEPARTMENTS values ('3',null,'4')
insert into DEPARTMENTS values ('5',null,'6')
insert into EMPLOYEES values ('7',null,'8','50001',null)
insert into DEPARTMENTS values ('1',null,'21240')
insert into DEPARTMENTS values ('-92279',null,'3')
insert into EMPLOYEES values ('4',null,'5','-97560',null)
insert into DEPARTMENTS values ('3',null,'4')
insert into DEPARTMENTS values ('5',null,'6')
insert into EMPLOYEES values ('7',null,'8','50000',null)
```
- 9 test data row(s) generated.** A green status message at the bottom of the right panel.

At the bottom of the interface, there's a footer with 'XData IIT Bombay · Automated SQL Grading' on the left and '© 2026 — built for learning, query by query.' on the right.

Figure 16: A successful generation on a custom schema. Z3 produced boundary values (50000 / 50001) that discriminate the query from its mutants.

7.1 Mutation types

Each ticked **mutation type** adds one targeted “killing” dataset aimed at a specific class of mistake — *Selection* (WHERE), *Joins (equi / non-equi)*, *Aggregation*, *Distinct*, *Group By* and *Having*. Tick only the types relevant to your query: a type that does not apply (e.g. *Aggregation* for a query without a GROUP BY) simply adds no rows. Toggling a type changes the output — for the salary query above, enabling *Selection* adds the boundary-killing rows (9 rows total), while disabling it falls back to the base dataset (3 rows).

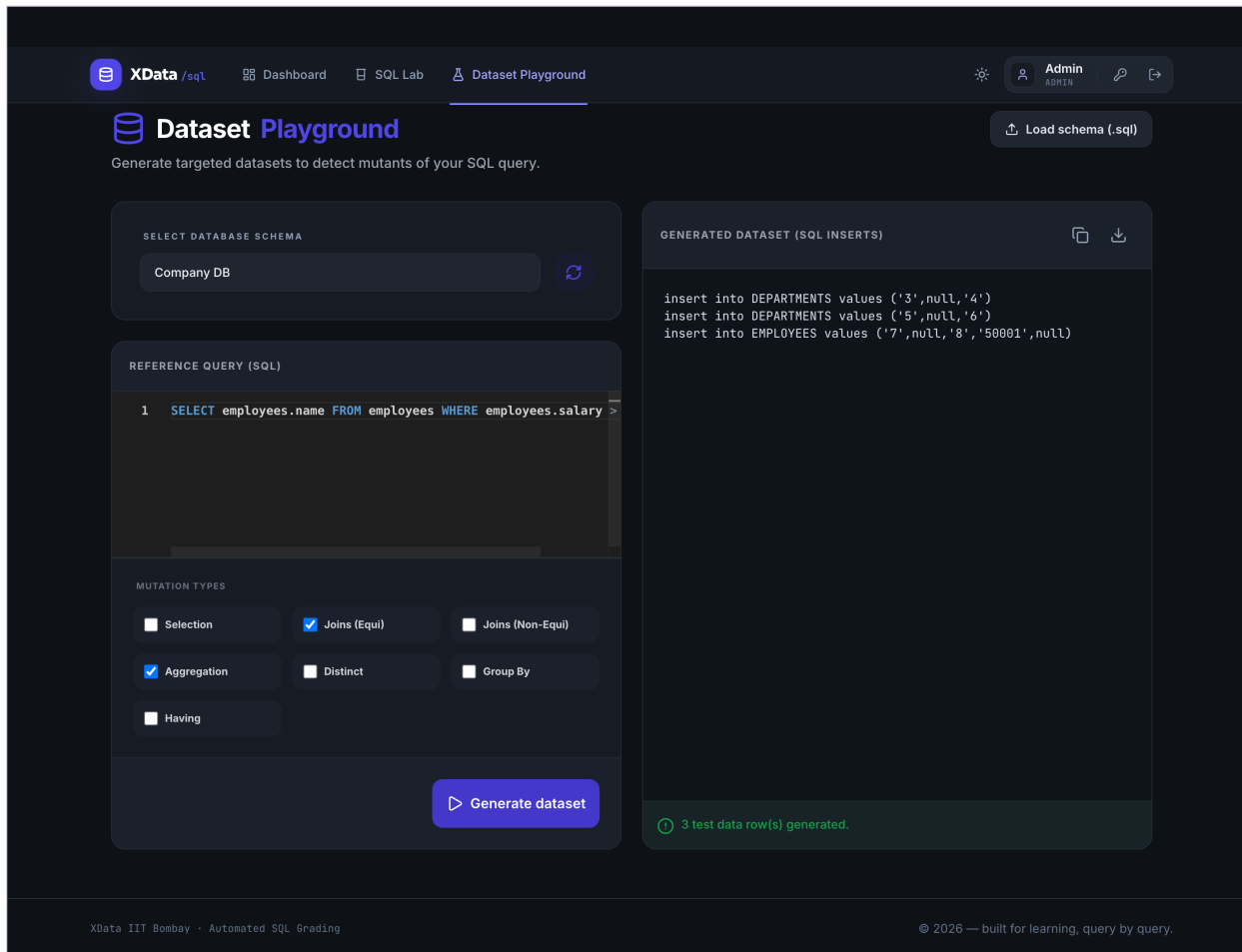


Figure 17: Mutation types are applied live: unticking a type removes its targeted rows from the generated dataset.

8 Tips & troubleshooting

- **Light / dark mode** — toggle with the sun/moon icon in the top bar.
- **Change password** — the key icon in the top bar.
- **“No assignment / course selected”** — some panels (Statistics, Schema management) need you to pick a course or assignment first; this is expected.
- **Ports** — the app uses 80 (frontend), 8080 (backend) and 5433 (database). Make sure they are free.
- **Fresh start** — `docker compose down -v` wipes the database and re-seeds the `admin1` account on the next start.

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